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Insights into the Dual Role of Lithium Difluoro(oxalato)borate Additive in Improving the Electrochemical Performance of NMC811||Graphite Cells

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KEYWORDS: Ni-rich layered oxide； NMC811； lithium-ion battery； solid electrolyte interphase； lithium difluoro(oxalato)borate; electrolyte additive;

ABSTRACT

Ni-rich layered oxides ($LiNi_xMn_yCo_zO_2$, $x \geq 0.6$, x+y+z=1) are promising positive electrode materials for high energy density lithium-ion batteries thanks to their high specific capacity. However, large-scale application of Ni-rich layered oxides was hindered by its poor structural and interfacial stability, especially during cycling at a high cut-off potential (i.e., ≥ 4.3 V, versus Li^+/Li). Herein, we demonstrated that lithium difluoro(oxalato)borate (LiDFOB) as a film forming additive plays a dual role on the electrode | electrolyte interphase formation in a $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$||graphite cell, meaning it can not only be reduced on the graphite negative electrode but also oxidized on the nickel-rich oxide $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$ positive electrode cycled at a high cut-off potential (4.4 V, versus Li^+/Li) prior to typical carbonate-based electrolyte constituents. As a result, the addition of 1.5 wt% LiDFOB greatly reduces the polarization and improves the cycling stability of the $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$||graphite cell, which shows a high discharge capacity of 198 mA h g^{-1} and more than 83.1% of the initial capacity was retained after 200 cycles at C/3 (the capacity retention obtained at the same cycling condition is only 59.9% for the cell without LiDFOB additive). Furthermore, the employ of LiDFOB additive also significantly suppresses the self-discharge of the $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$||Li cell during high temperature and long-term room temperature storage at 4.4 V. These electrochemical performance enhancements could be attributed to the participation of LiDFOB in forming a stable and Li^+ transfer favorable protective

layer that is rich in inorganic boron-, fluorine-, and carbonate compounds on both the surface of the $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$ positive electrode and the graphite negative electrode, thus suppressing the electrolyte decomposition on the positive electrode and negative electrode surfaces and decreasing the dissolution of transition metal ions from the positive electrode bulk.

1. Introduction

Lithium-ion batteries (LIBs) are widely employed in portable electrical devices, electrical vehicle, and large-scale energy storage systems and are subject to on-going modification to meet the growing demands for higher energy and power densities.[1] In recent years, nickel-rich layered oxides ($LiNi_xMn_yCo_zO_2$, $x \geq 0.6$, $x+y+z=1$), which has a lower cost and can offer higher specific capacity (i.e., 215 mAh g^{-1} when charged to 4.4 V versus Li^+/Li) compared to $LiCoO_2$, is becoming an attractive positive electrode material for the development of high energy density LIBs.[2-4] However, decomposition of electrolyte on the positive electrode surface catalysed by the highly oxidative Ni^{3+}/Ni^{4+} is becoming a major cause leading to capacity fading of batteries, hindering the large-scale application of nickel-rich oxides.[5-7]

Solutions to address the unstable interface problem of nickel-rich oxides mainly include decreasing the surface area of the oxide particles, replacing the carbonate solvents with other stable electrolytes (e.g., ionic liquids, nitriles),[8,9] developing a stable coating layer on the oxide particle surface during material synthesis,[10,11] and employing film-forming electrolyte additives for in situ formation of a cathode electrolyte interphase

(CEI) layer.[12-15] Among these solutions, forming a protective CEI layer on the nickel-rich oxide by film-forming additives is found to be an effective strategy to suppress the oxidative decomposition of electrolyte thus improving the cyclability.[2,13,14] Especially, considering the tremendous successes of additives in forming solid electrolyte interphase (SEI) on the negative electrode surfaces on suppressing the reductive decomposition of carbonate electrolytes,[5,16-18] introducing additives to form a CEI layer may also play a critical role in improving the cyclability of advanced positive electrode active materials such as nickel-rich oxides.

Lithium difluoro(oxalato)borate (LiDFOB) is one of novel CEI-forming additives and it can effectively improve the high-voltage and high-temperature performance of positive electrode materials.[19-22] For example, Zonghai Chen et al. demonstrated that the LiDFOB loading in a carbonate electrolyte greatly influences the performance of $LiNi_{0.6}Mn_{0.2}Co_{0.2}O_2$||Li cells and showed improved capacity retention at 2.8–4.5 V when the electrolyte contains 0.5 wt% LiDFOB.[19] Lee et al. found that LiDFOB additive in LiFSI electrolytes is outstanding as a corrosion inhibitor for protecting the Al current collector of a $LiCoO_2$||graphite cell since it helps to form a passive layer composed of Al_2O_3, Al-F, and B-O species.[20] Choi et al. reported that adding LiDFOB to a carbonate electrolyte greatly improve the electrochemical stability of $Li_{1.17}Ni_{0.17}Mn_{0.5}Co_{0.17}O_2$||graphite cells.[23]

In this work, we report using LiDFOB as an electrolyte additive to improve the cyclability of a $LiNi_{0.83}Mn_{0.05}Co_{0.12}O_2$ (NMC811)||graphite cell under high cut-off voltage.

The mechanisms of how the additives affect the battery performance is investigated in detail.

2. Results and Discussion

2.1 Evidence of LiDFOB in film-formation

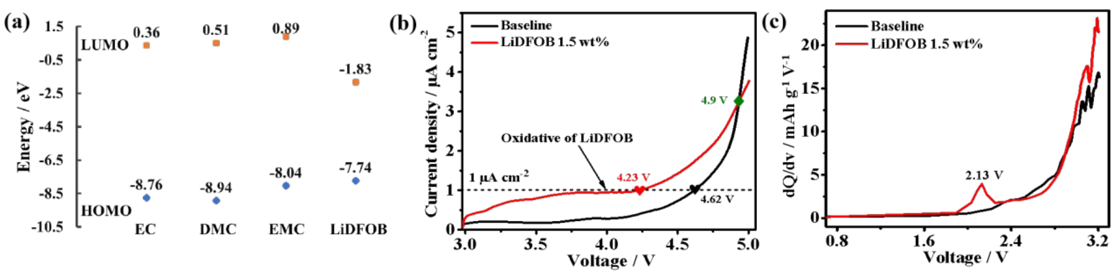

Figure 1. (a) Calculated HOMO and LUMO energies of EMC, EC, DMC, and LiDFOB. **(b)** LSV results recorded at stainless-steel plate||Li coin cells in the baseline electrolyte and 1.5 wt% LiDFOB–containing electrolytes at a scan rate of 1 mV s^{-1}, a stainless-steel plate was used as the working electrode and a Li metal foil was used as counter electrode. **(c)** differential capacity profiles of the first charge (between 0.8-3.2 V) of NMC811||graphite coin cells with/without LiDFOB additive.

The energy of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) are used to estimate the oxidation and reduction potential of an additive molecule in screening of its film-forming properties. Usually, if the HOMO level of the additive is higher than that of the electrolyte solvent, the additive is preferentially oxidized before the solvent to form an CEI on positive electrode; while additives with LUMO levels lower than that of the electrolyte solvent are preferentially reduced to form a SEI on the surface of the negative electrode.[24] As shown in **Figure 1a,** the HOMO and LUMO energy of the LiDFOB additive and the EC, EMC, and DMC solvents was calculated according to the reported method.[25] LiDFOB has a higher HOMO and lower LUMO energy levels than that of the EC, EMC, and DMC, thereby it can be both oxidized on the positive electrode surface at a lower potential and reduced on the negative electrode surface at a higher potential than the above solvents, playing a dual role on the electrode | electrolyte interphase formation. Based on the calculated HOMO and LUMO levels, electron affinities and electron dissociation energies of solvents and the LiDFOB additive were also calculated and showed in **Figure S1**.

The preferential oxidation of LiDFOB was verified by linear sweep voltammetry (LSV) measurement obtained from a stainless-steel plate||Li coin cell in electrolyte with/without LiDFOB additive. As shown in **Figure 1b,** below 4.9 V, the oxidation current of the cell with addition of 1.5 wt% LiDFOB is much larger than that of the baseline, implying the oxidation of the LiDFOB-containing electrolyte starts before the baseline. Interestingly, when the voltage rises to 4.9 V, the oxidation current of the the baseline cell increases sharply and becomes far exceeding that of the LiDFOB-containing electrolyte, indicating that the deposited products formed on the electrode surface from the oxidation of LiDFOB at lower potential can suppress the severe oxidation decomposition of the organic solvents at a high potential. The reduction of LiDFOB was confirmed by the differential capacity profiles (between 0.8-3.2 V) during the initial charge of

NMC811||graphite coin cells in electrolyte with/without LiDFOB additive. As shown in **Figure 1c**, a clear peak is observed at 2.13 V for the cell charged in the electrolyte containing 1.5 wt% LiDFOB but not at the cell using the baseline electrolyte, indicating the reduction of LiDFOB at the graphite negative electrode. In summary, theoretical calculations and experimental results revealed that LiDFOB can be both reduced on the negative electrode surface and oxidized on the positive electrode surface prior to the carbonate solvents to form CEI/SEI that may protect the positive electrode and the negative electrode.

2.2 Effects of LiDFOB on electrochemical behaviour of NMC811||Li metal cells

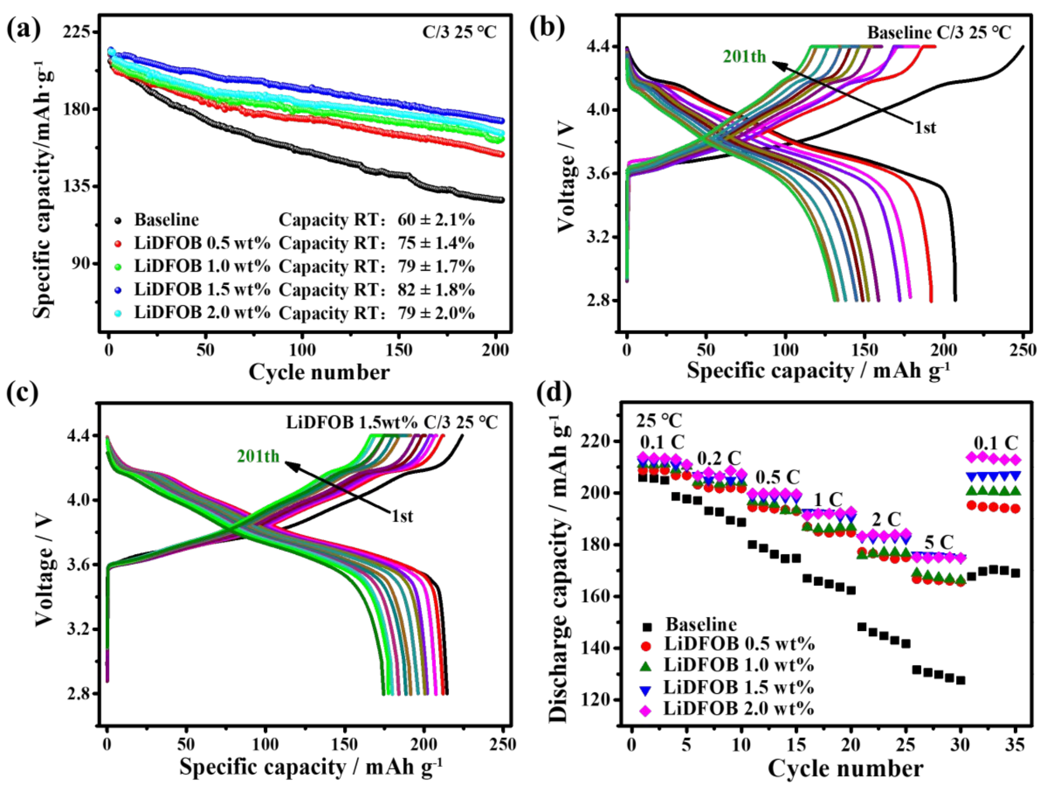

Figure 2. Electrochemical characterization of NMC811||Li metal cells in electrolytes with different amount of the LiDFOB additive. **(a)** cycling performance at C/3 after two cycles at 0.1 C (1 C = 180 mA g^{-1}), **(b-c)** voltage profile evolution of the cells cycled in the blank and 1.5 wt% LiDFOB-containing electrolyte, **(d)** rate capability of NMC811|| Li metal cells in electrolytes with different amount of LiDFOB additive.

The effect of the LiDFOB additive on the NMC811 positive electrode was evaluated in NMC811||Li metal cells cycled between 2.8–4.4 V (**Figure 2**). All cells were first cycled between 2.8–4.4 V at 0.1 C for the first two cycles at room temperature as a SEI/CEI formation process. The cycling performances of NMC811||Li cells in electrolytes with various LiDFOB contents were compared at C/3 (**Figure 2a**) and 1C (**Figure S2a**). Obvious electrode capacity fading was observed for the NMC811||Li cell cycled in the baseline electrolyte, resulting in a capacity retention of only 60 ± 2.1% after 200 cycles at C/3. While the capacity retention for the NMC811||Li cells cycled in electrolytes containing 0.5, 1, 1.5, and 2 wt% LiDFOB are 75 ± 1.4%, 79 ± 1.7%, 82 ± 1.8%, and 79 ± 2.0%, respectively after 200 cycles. The cyclability of the NMC811||Li cells with different amount of LiDFOB measured at 1 C shows similar results (**Figure S2a**). In both the C/3 and the 1C cycling test, the cell cycled in electrolytes containing 1.5 wt% LiDFOB exhibited the highest capacity retention.[25] **Figure S2b** presents the Coulombic efficiency (CE) of NMC811||Li cells using different electrolytes during cycling between 2.8–4.4 V at 0.1C for 2 cycles followed by 1C. The cell cycled in the baseline case electrolyte shows a first cycle CE of only 81±3.2%, while that of the cells cycled in the electrolyte containing LiDFOB additive is much higher and a maximum value (94±1.6%) is obtained for the cell cycled in the electrolyte containing 1.5 wt% LiDFOB. The high CE of the LiDFOB-containing cell could be due to that the intensive reactivity of NCM811 with electrolyte was greatly suppressed by the CEI and SEI formed with the help of LiDFOB, and the strict environment control of the cell preparation process, including mixing, coating, rolling, weighing of the electrode in a drying room (dew point: -50 ℃), and drying the electrode in an vacuum oven before assembling the cells in an Ar-filled glove box. The CE of the cells cycled in LiDFOB-containing electrolyte during cycling is also higher and more stable than that of the controlled group during long-term cycling. These improved CE results could be explained by that the CEI and SEI formed on the surface of the high potential Ni-rich positive electrode and the Li negative electrode, respectively, at the presence of the LiDFOB additive in the first charge prevent further electrolyte decomposition.

The voltage profile evolution of the baseline cell and the 1.5 wt% LiDFOB-containing cell in **Figure 2b** and **c** showing that the polarization of the former cell increased with the cycling, while that of the latter cell kept nearly constant, indicating that addition of the LiDFOB can stabilize the cell from impedance growth during cycling probably by supressing decomposition of the electrolyte, which also agrees with the CE results (**Figure S2b**). **Figure 2d** shows the discharge capacity of NMC811||Li cells cycled in the range of 2.8-4.4 V under different rates and reveals that electrolytes containing 2.0 and

1.5 wt% LiDFOB exhibit best rate capabilities. Furthermore, the capacity of baseline NMC811||Li cells at 0.1C could not be recovered after cycled at 5 C; in contrast, almost all the 0.1C capacity can be recovered after cycled at 5 C when the electrolyte contained 2.0 or 1.5 wt% LiDFOB. These results reveal that CEI formed at the presence of the LiDFOB additive enable faster Li^+ transportation on Ni-rich positive electrode.

Since the LiDFOB is supposed to be reduced on the Li negative electrode prior to the carbonate solvents to form a SEI layer on the Li negative electrode surface which could improve the electrochemical performance of the NMC811||Li cells, we have also investigated the effect of LiDFOB on the Li||Li symmetric cells via monitoring the overpotential of the Li plating/stripping cycling. As shown in **Figure S3a–e,** when the Li||Li symmetric cells was cycled at a current density of 0.25 mA cm^{-2} (same value as that of the NMC811||Li cells cycling at C/3), no significant difference can be observed on the Li striping/plating overpotential for electrolytes containing 0, 0.5, 1, 1.5, and 2 wt% LiDFOB. These indicate that the addition of LiDFOB does not show positive effect on the Li striping/plating at the negative electrode, which could be caused by the large volume change of the Li electrode that reduce the stability of the LiDFOB-induced SEI. Therefore, the electrochemical performance enhancement of the NMC811||Li cells with LiDFOB additive could mainly be origined from the LiDFOB-induced CEI on NMC811 positive electrode. Since electrolyte with 1.5 wt% of LiDFOB showed the greatest improvement on the electrochemical performance, it was used to demonstrated the effect of LiDFOB in the following section.

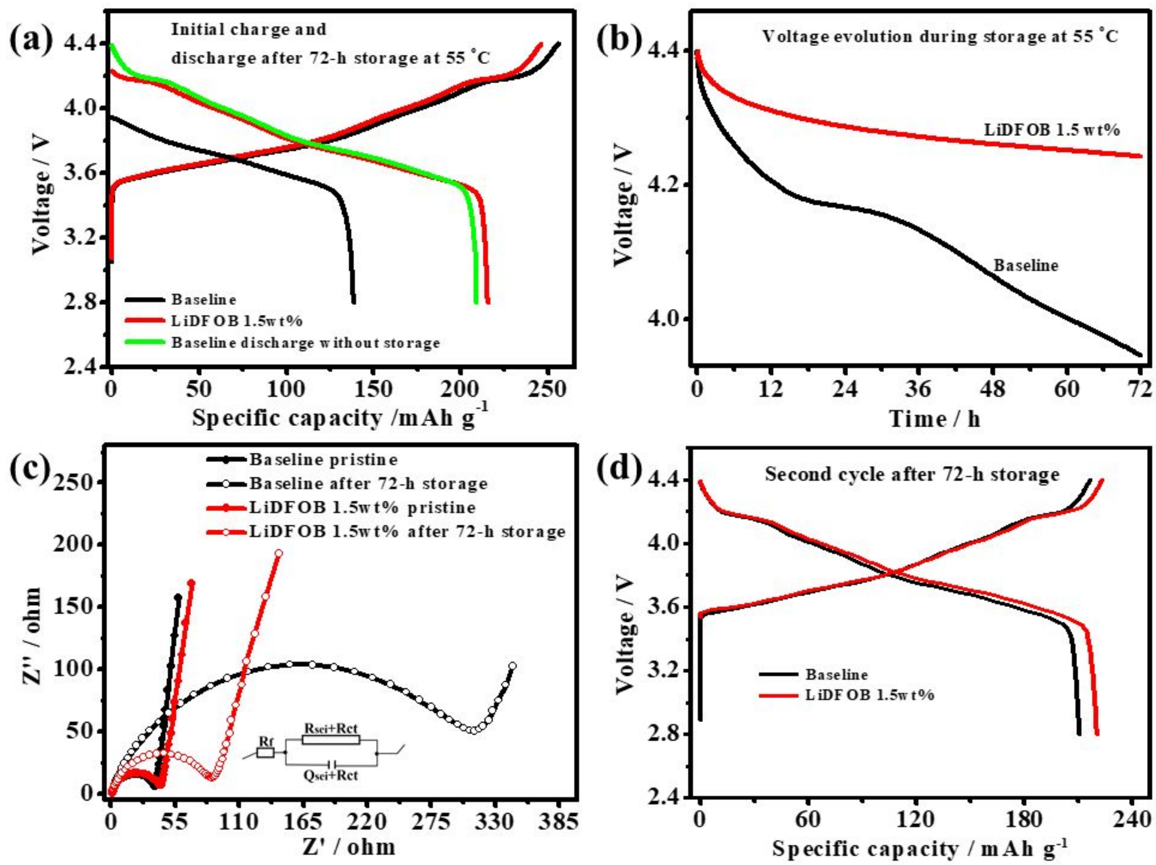

Figure 3. Self-discharge tests performed at 55 °C for NMC811||Li cells containing different electrolytes, **(a)** voltage profiles of the initial charge and discharge after 72 hrs storage at 55 ℃, **(b)** OCV evolution during 72 hrs rest, and **(c)** EIS spectra recorded before and after the self-discharge test**, (d)** voltage profiles of the second cycle after 72 hrs storage at 55 ℃.

Self-discharge behaviour of NMC811||Li cells during storage at high temperature (4.4 V@55 °C for 72 hrs) and room temperature (4.5 V @25 °C for 400 hrs) were also investigated to evaluate the effect of the LiDFOB additive. **Figure 3a** shows the charging profile at 55 °C at 0.1 C to 4.4 V and **Figure 3b** records the variations of open-circuit voltage (OCV) of the cells during storage at 55 °C for 72 hrs, and the results show that the cell with baseline electrolyte suffered from a rapid OCV drop from 4.4 V to 3.95 V while the OCV of the cell with LiDFOB additive is slowly decreasing to 4.16 V in the 72 hrs duration. As a result, the NMC811||Li cell with LiDFOB additive showed a capacity of 214 mAh g^{-1} after the 72 hrs storage at 55 °C but that without additive only delivered 139 mAh g^{-1} at the same testing condition (**Figure 3a**). It is also worth noting that without storage, the NMC811||Li cell without additive can delivery 209 mAh g^{-1} (green line in **Figure 3a**), indicating significant capacity loss during the storage. EIS results in **Figure 3c** revealed that during the high temperature storage the impedance of the cell using the additive-free electrolyte increased largely compared to the cell with LiDFOB additive, indicating formation of high Li^{+} transfer resistant CEI from decomposition of electrolyte. Self-discharge behaviour of NMC811||Li cells charged to 4.5 V for 400 hrs was also evaluated at during storage at room temperature. The OCV of the cell with LiDFOB additive is quite stable while that with baseline electrolyte first drop sharply then the speed slow down but keep decreasing to 3.69 V (**Figure S4**), resulting a large capacity loss (only 62 mAh g^{-1} was obtained after the storage, which is much smaller than that (205 mAh g^{-1}) of the LiDFOB-containing cell). These self-discharge results prove that the LiDFOB-induced CEI during the first charge process is helpful to prevent side reactions during high temperature and high voltage storage. Surprisingly, most the capacity loss of the baseline cell during self-discharging test can be recovered in further cycling (**Figure 3d**), indicating that the capacity loss of the baseline cell during the high temperature storage mainly caused by reaction with electrolyte, but not due to irreversible structure transformation of the NMC811 materials.

2.3 Electrode characterization

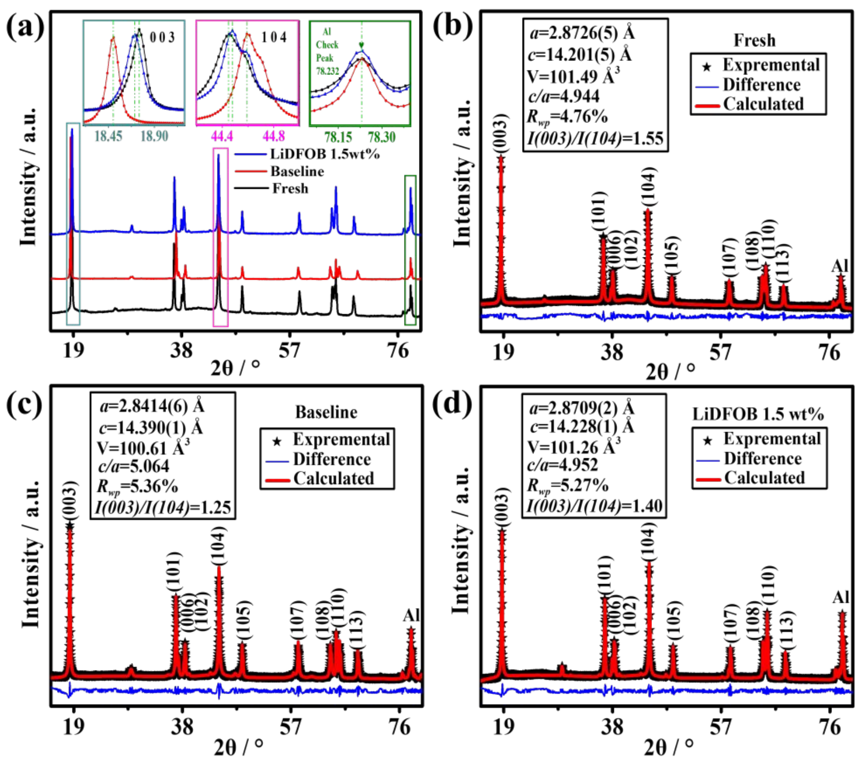

Figure 4. (a) XRD patterns of NMC811 electrodes that are fresh and disassembled from fully discharged NMC811||Li cells after 200 cycles at C/3 C in baseline and 1.5 wt% LiDFOB-containing electrolytes. Contents of transition metals deposited on surface of lithium electrodes from NMC811||Li cells using electrolyte without and with 1.5 wt% LiDFOB additive after 200 cycles at C/3. XRD Rietveld refinements of **(b)** fresh NMC811 electrodes and those subjected to 200 cycles at C/3 in **(c)** baseline and **(d)** 1.5 wt% LiDFOB–containing electrolytes.

Figure 4a presents the X-ray diffraction patterns (XRD) of fresh NMC811 electrodes and those disassembled from fully discharged NMC811||Li cells after 200 cycles at 1 C in electrolytes without and with 1.5 wt% LiDFOB additive. Both pristine and cycled electrodes exhibited strong characteristic NMC peaks that could be indexed to (003), (101), (006), (102), (104), (105), (107), (108), and (110) reflections. The diffraction peak of Al from the current collector at 2theta of 78.232° was used as a reference to examine the 2theta shifting of the (003) and (104) peaks that belong to the NMC811 material. As shown in the insert parts in **Figure 4a**, the (003) peak is obviously shifting to a lower 2theta angle and the (104) peak to a higher 2theta angle for the baseline positive electrode compared to the fresh electrode, indicating an expanding of the c axis and shrinkage of the a/b axis probably due to loss of lithium during cycling, whereas no such shift was observed when a LiDFOB-containing electrolyte was employed. This difference in unit cell parameters evolution during cycling can also be verified by Rietveld refinement results showed in **Figure 4b–d** that used a layered α-$NaFeO_2$ (R-3m space group) as a calculation model. In addition, the (003)/(104) peak intensity ratio of the electrode cycled in the additive-free electrolyte decrease largely (1.55 to 1.25) compared to that (1.55 to 1.40) of the electrode with LiDFOB additive, indicating that large amount of Ni diffused into the Li layer which could cause sluggish Li ion conducting. Furthermore, Inductive coupled plasma mass spectrometry (ICP-MS) testing results of the lithium electrode of the NMC811||Li cells cycled in baseline electrolyte featured a much higher transition metal content than that cycled in the 1.5 wt% LiDFOB–containing electrolyte (**Figure**

S5a). X-ray photoelectron spectroscopy (XPS) analysis also found fine Ni 2p spectrum on the lithium electrode cycled in the baseline electrolyte, whereas no Ni can be detected in that cycled in 1.5 wt% LiDFOB–containing electrolyte (**Figure S5b** and **c**).[26,27]

These characterization results of positive electrode structure and transitional metal evidence on the negative electrode reveal that LiDFOB additive is helping to form a CEI layer on the NMC811 positive electrode surface, preventing continues decomposition of electrolyte during long-term cycling, and supressing transitional metal dissolution from the NMC structure, resulting a stable interface and good electrochemical performance.

2.4 Electrochemical behaviour of NMC811||graphite full cell

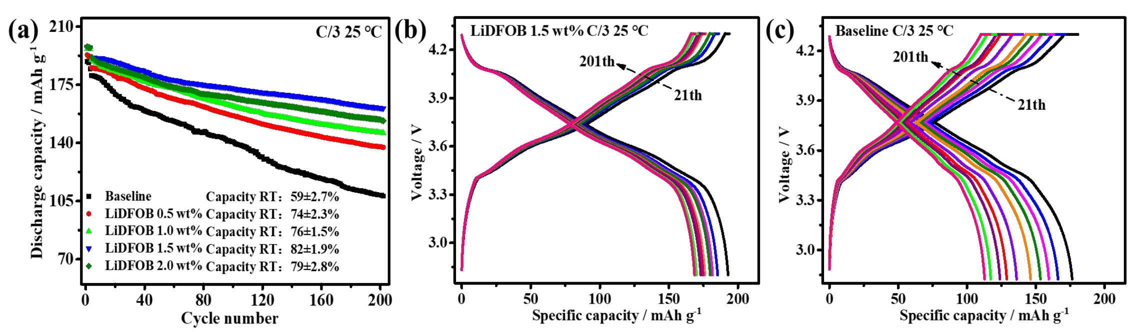

Figure 5. Electrochemical performance of the NMC811||graphite cells **(a)** cycling performance and **(b-c)** voltage profiles of the cell cycling in blank and 1.5 wt% LiDFOB-containing electrolyte under C/3 between 2.8–4.3 V at room temperature.

Since the low LUMO of the LiDFOB also imply it can easier be reduced than the carbonate electrolyte solvents on the negative electrode surface, NMC811||graphite full cells were assembled to study the positive electrode and negative electrode interface together. As shown in **Figure 5a**, the cycling performance is the best in the case of 1.5 wt% LiDFOB, and the worst in the additive-free electrolyte. This trend is similar to that of the NMC811||Li cells, but the capacity retention is slightly higher in the full cell after 200 cycles at C/3 (82 $\pm$ 1.9% and 82 $\pm$ 1.8% for the Li metal cell and full cell, respectively), which could be attributed to the contribution of the graphite negative electrode. **Figure 5b** and **c** showed the voltage profile evolution of the baseline cell and the 1.5 wt% LiDFOB-containing cell. The increase of polarization with cycling as seen in the baseline cell was largely supressed in the 1.5 wt% LiDFOB-containing cell, and the overpotential of the NMC811||graphite full cell during cycling is also much more stable than that of the NMC811||Li metal cell, which could be attributed to the more stable LiDFOB-induced SEI on the graphite negative electrode than the Li electrode.

2.5 Characterization of CEI and SEI in the NMC811||graphite cell

The morphology of the LiDFOB induced SEI/CEI layers in the NMC811||graphite full battery after 200 cycles was observed by scanning electron microscopy (SEM) and transmission electron microscopy (TEM). **Figure S6** presents the surface topography of positive and negative electrodes of the NMC811||graphite cells that cycled in electrolytes without and with 1.5 wt% LiDFOB additive for 200 cycles, comparing to fresh electrodes. **Figure S6a-c** show that the fresh NMC811 electrode and the positive electrode cycled in the baseline electrolyte has a relatively clean surface, while a thin deposited layer is found on the surface of the positive electrode that cycled in the electrolyte with 1.5 wt% LiDFOB additive (**Figure S6c**). High-resolution TEM imaging found that (i) the surface of fresh particles (**Figure S6d**) was not covered by any film and (ii) the CEI obtained in the 1.5 wt% LiDFOB–containing electrolyte **(Figure S6f)** was much thicker and more uniform than that in the baseline electrolyte (**Figure S6e**). Thus, the CEI produced in the 1.5 wt% LiDFOB–containing electrolyte should be better suited for inhibiting electrolyte decomposition and preventing metal ion elution. **Figure S6g-i** show that after 200 cycles at room temperature, the morphology of graphite negative electrode cycled in the electrolyte containing 1.5 wt% LiDFOB (**Figure S6i**) remains sharp edges as the fresh one and covered with some decomposed products, while the negative electrode that was cycled in the baseline electrolyte is covered with a very thick layer (**Figure S6g-h**), which could be ascribed to continuous decomposition of electrolyte on the graphite negative electrode during the cycling.

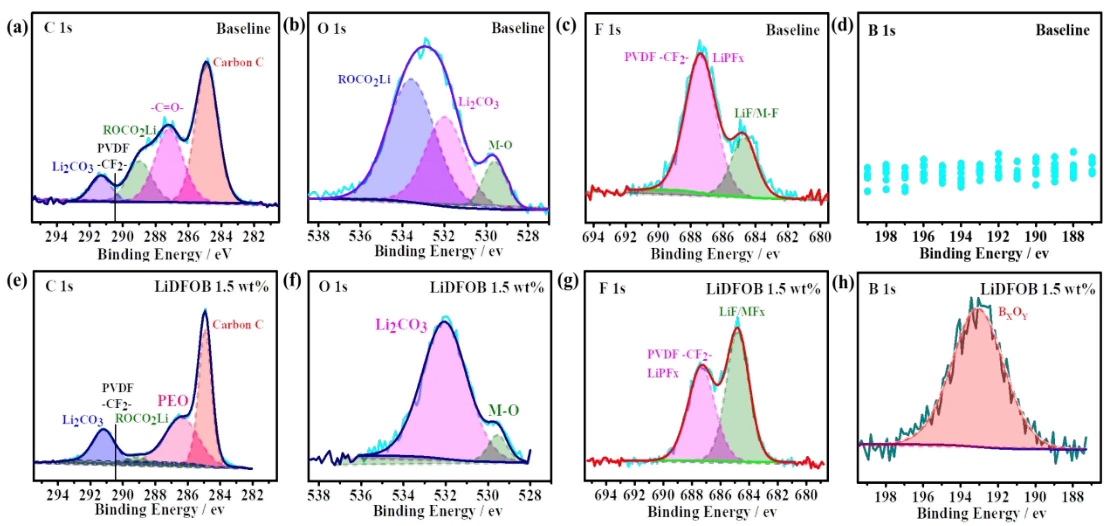

Figure 6. (a, e) C 1s, **(b, f)** O 1s, **(c, g)** F 1s, **(d, h)** B 1s XPS spectra of NMC811 positive electrodes cycled in the baseline and the 1.5 wt% LiDFOB–containing electrolytes, respectively, the positive electrodes are disassembled from fully discharged

NMC811||graphite cells after 200 cycles at C/3. All charge–discharge were measured between 2.8–4.3 V at room temperature.

XPS measurement was used to explore the chemical constituents of the CEI/SEI layer from the NMC811||graphite cells cycled with and without the LiDFOB additive. **Figure S7a** compares the elemental compositions of CEI, showing that compared to that cycled in the baseline electrolyte, the CEI produced in the presence of 1.5 wt% LiDFOB featured obvious B (5.08) and slightly higher F contents (36.74 vs. 29.97 at%), lower C and P contents (44.22 vs. 52.74 at% and 1.88 vs. 3.61 at%), and almost identical O content. The P element might be mainly come from the decomposition products of $LiPF_6$, indicating that the addition of LiDFOB reduces the $LiPF_6$ electrolyte salt decomposition. XPS fine spectra in **Figure 6a–h** provides exhaustive data on the CEI composition. Compared to the baseline positive electrode, that cycled in the 1.5 wt% LiDFOB-containing electrolyte featured a lower content of C–C bonds (284.8 eV, originated from the carbon black) and a higher content of PEO-like moieties (286.5 eV) (**Figure 6a** and **e**), which is believed to be a good ionic conductor.[28,29] The fine O 1s spectrum (**Figure 6b** and **f**) clearly indicated that the LiDFOB-containing electrolyte CEI contained more stable Li_2CO_3, while the CEI produced in the presence of baseline-electrolyte contained more $ROCO_2Li$, which is metastable.[30] Additionally, using the content of O–M bonds (529.5 eV) as an indicator of CEI thickness,[31] we found that the baseline-electrolyte CEI was thinner than that produced in the presence of LiDFOB, in agreement with the results of TEM imaging. The F 1s spectra revealed that the CEI produced in the presence of LiDFOB featured an increased content of LiF/MF_x species (684.5–685.5 eV) which could originate form decomposition of the LiDFOB additive (**Figure 6c** and **g**). Finally, the fact that a B–O peak (B_xO_y species) at 193.0 eV was present in the sample cycled in the 1.5 wt% LiDFOB-containing electrolyte (**Figure 6h**) but not in that cycled in the baseline electrolyte (**Figure 6d**) confirmed that CEI formation involves the oxidation of $DFOB^-$.[31,32]

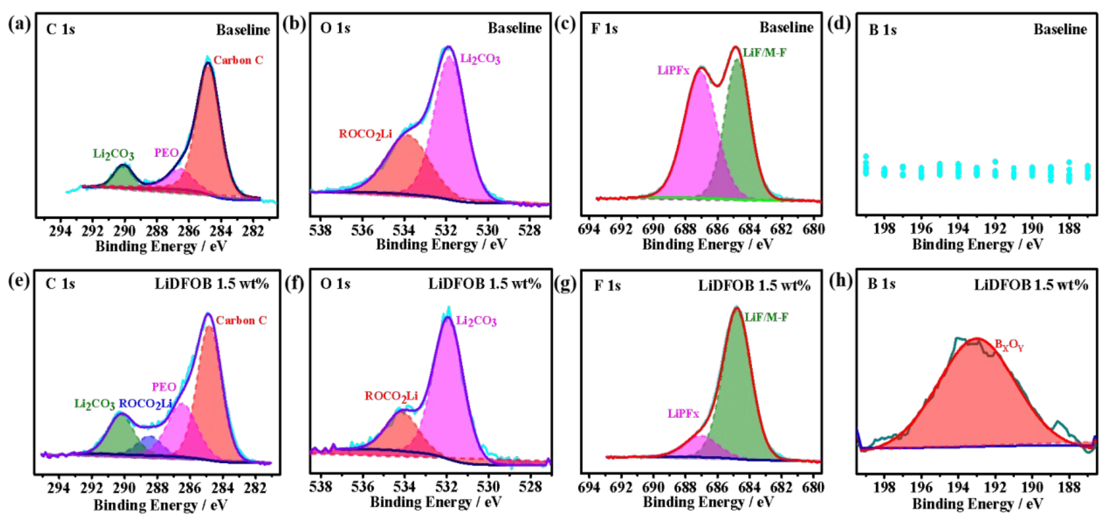

Figure 7 (a, e) C 1s, **(b, f)** O 1s, **(c, g)** F 1s, **(d, h)** B 1s XPS spectra of negative electrodes disassembled from fully discharged NMC811||graphite cells after 200 cycles at C/3 (1 C = 180 mA g^{-1}) in electrolyte without and with 1.5 wt% LiDFOB additive. All charge–discharge measurements were conducted in the range of 2.8–4.3 V at room temperature.

Figure S7b compares the elemental compositions of SEIs obtained from the negative electrode with different electrolytes, showing that SEI formed in the presence of 1.5 wt% LiDFOB featured much higher F contents (19.03 vs. 6.44 at%), lower Li and P contents (25.02 vs. 57.47 at% and 1.35 vs. 5.09 at%), an obvious B signal. XPS fine spectra of the SEI layer were shown in **Figure 7a-h**. Similar to that observed at the positive electrode, compared to that cycled in the baseline electrolyte the graphite negative electrode cycled in LiDFOB-containing electrolyte contains more Li_2CO_3 (C1s, 290.5 eV), PEO-like species (C1s 286.5 eV) and LiF (F1s 684.8 eV), and evident B–O peak (B_xO_y species, B1s 192 eV) , but obvious less $ROCO_2Li$ (O1s 533.8 eV) and $LiPF_x$ species (F1s 687 eV) that decomposed from the $LiPF_6$ salt.

Even though we can observe B_xO_y species on both the positive electrode and negative electrode, the peak shape of the two B1s spectra behave a bit different. However, it is difficult to obtain more detail information for the B_xO_y species from the XPS B1s spectra, more detail investigation on the decomposition mechanism of LiDFOB on the positive and negative electrode needed to be conducted in future.

In view of the above results, it can be concluded that the LiDFOB additive helps to form CEI and SEI, that rich in stable inorganic boron-, fluorine- and carbonate compounds, but lack of $LiPF_x$ species and organic compounds like lithium alkyl carbonate, on the surface of the NMC811 positive and graphite negative electrode.

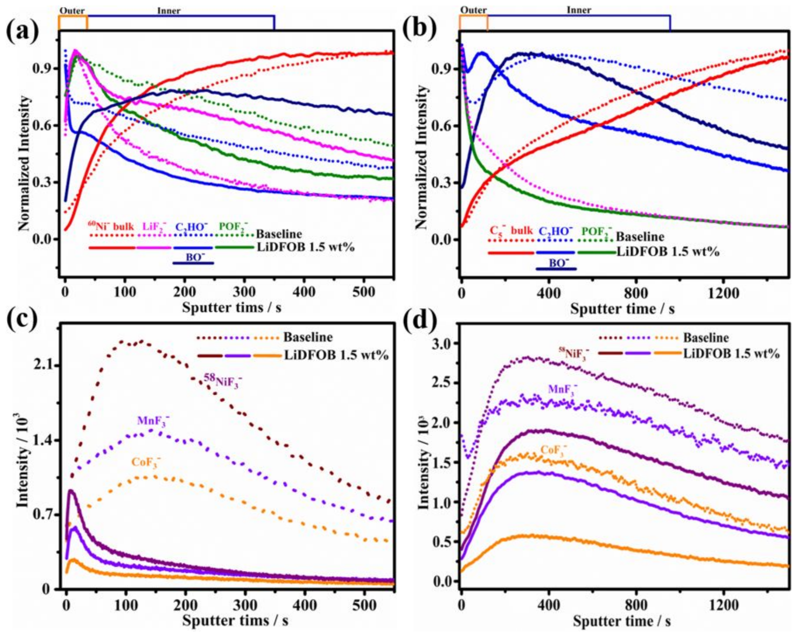

Figure 8. TOF-SIMS depth profiles of selected fragments obtained on NMC811 positive electrodes **(a)** and graphite negative electrodes **(b)** that disassembled from full cells after 200 cycles in electrolyte without and with 1.5 wt% LiDFOB additive. TOF-SIMS depth profiles of the deposited transition metal fragments acquired on NMC811 positive electrodes **(c)** and graphite negative electrodes **(d)** that disassembled from full cells after 200 cycles in electrolyte without and with 1.5 wt% LiDFOB additive.

Time of flight secondary ion mass spectrometry (TOF-SIMS), which is a highly surface-sensitive technique with ultrahigh chemical sensitivity,[31] was applied to further analysis of the CEI and SEI chemistry. **Figure 8a-d** presents the depth profiles of selected secondary-ion fragments obtained from the NMC811 electrodes and graphite negative electrodes that cycled in electrolyte without and with 1.5 wt% LiDFOB additive, including $^{60}Ni^{-}$ (representing bulk NMC811), C_5^{-} (representing carbon species from the bulk graphite), C_2HO^- (representing organic species related to the carbonate electrolyte decomposition), POF_2^- and PO_2^- (representing inorganic constituents related to $LiPF_6$ decomposition), BO^- (representing deposited species from LiDFOB decomposition), LiF_2^- (representing insoluble species from LiDFOB and $LiPF_6$ decomposition), and $^{58}NiF_3^-/CoF_3^-/MnF_3^-$ (representing transition-metal dissolution).[31,34-37] **Figure 8a** and **b** showed that the chemical composition of both CEI and SEI formed in electrolyte without and with 1.5 wt% LiDFOB additive can be roughly recognized as a "double-layer" structure. Specifically, as shown in **Figure 8a,** for the NMC811 cycled in the baseline electrolyte, LiF_2^- only externally present in the outer layer, whereas LiF_2^- exists on both the outer and inner (together with the BO^- species) layer of the CEI formed in the LiDFOB-containing electrolyte. Furthermore, compared to the CEI that formed in the baseline electrolyte, the intensities of C_2HO^- and POF_2^- in the inner layer of the CEI formed in the 1.5 wt% LiDFOB-containing electrolyte is much lower, indicating that at the early state of the cycling, the main decomposition components on the baseline electrolyte and the LiDFOB-containing electrolyte are carbonate solvent and LiDFOB, respectively. **Figure 8b** showed that the main component present in the inner layer of the SEI formed in the blank and 1.5 wt% LiDFOB-containing electrolyte are C_2HO^- and BO^-, corresponding to decomposition of carbonate solvent and LiDFOB, respectively. It was also worth noting that the POF_2^- mainly present on the outer layer of both the CEI and SEI, thus they could come from decomposition of residual $LiPF_6$ salt during sample preparation for TOF-SIMS measurement. As shown in **Figure 8c**, transition-metal dissolution is effectively reduced within CEI formed in 1.5 wt% LiDFOB-containing electrolyte, as evidenced by much weaker $^{58}NiF_3^-/MnF_3^-/CoF_3^-$ signals in the CEI layer and the intensities quickly decrease after sputtering for ~20 s from the top CEI surface, further verifying the protective role of LiDFOB-assisted CEI for the high nickel positive electrode. Meanwhile, as shown in **Figure 8d,** the deposition of dissolution transition metal ions on the surface of the graphite negative electrode cycled in the baseline electrolyte is also much higher than that cycled in the LiDFOB-containing electrolyte, consistent with the ICP-MS results. Since sputtering times for the emergence of stable $^{60}Ni^-/C_5^-$ signal and the disappearance of the CEI/SEI species were ~ 550 s for the positive electrodes and ~1500 s for the negative electrodes, the CEI should be much thinner than the SEI.

2.6 Electrolyte composition analysis

Many literatures mentioned that the EMC solvent may be decomposed into DMC, DEC, dimethyl 2,5-dioxahexane dicarboxylate (DMDOHC), ethylmethyl 2,5-dioxahexane dicarboxylate (EMDOHC), and diethyl 2,5-dioxahexane dicarboxylate (DEDOHC) on the NMC electrode during cycling due to a trans-esterification reaction, which may has some negative effects on the electrochemical performance of lithium-ion batteries.[38,39] Therefore, to confirm that LiDFOB can suppress the carbonate solvent decomposition, we analyzed the composition of the baseline and LiDFOB-containing electrolyte before and after 100 cycles at 1C using a Gas Chromatography-Mass Spectrometer (GC-MS). To ensure enough electrolyte volume extracted from the cycled cell is enough for the analysis, a Whatman GF/D separator was used to assemble 2025-type cells. **Figure S8a** and **b** clearly show that the carbonate solvent of the 1.5 wt% LiDFOB-containing electrolyte did not change after 100 cycles, while the relative peak intensity of EMC in the baseline electrolyte is getting weaker after 100 cycles, indicating that trans-esterification reaction happened to the EMC during cycling. These results indicate that LiDFOB can suppress the decomposition of carbonate solvents by forming CEI/SEI on the NMC811 positive electrode and the Li metal negative electrode.

3. Conclusion

A range of instrumental and electrochemical characterisation techniques were used to explore the effects of LiDFOB addition on NMC811||Li and NMC811||graphite cell performance and explain the mechanism of LiDFOB-involved film formation. It was demonstrated that the addition of LiDFOB additive can improve the electrochemical performance of NMC811||graphite cells due to it is critical role on the formation of CEI and SEI that rich in stable inorganic boron-, fluorine- and carbonate compounds, but lack of salt decomposed species such as $LiPF_x$ and organic compounds like lithium alkyl carbonate, on the surface of the NMC811 positive electrode and graphite negative electrode, imply that SEI/CEI formed on the presence of LiDFOB additive figure low Li^+ transfer impedance and are stable to supress the decomposition of electrolyte during cycling. We believe the in-depth understanding on how film-forming additive affect the CEI/SEI thus the cell performance presented in this study will provide inspiration for future development of CEI film-forming additive for nickel-rich oxides and accelerate their large-scale application in high energy density LIBs.

4. Experimental

4.1 Chemicals and reagents

The baseline electrolyte (i.e., 1 M $LiPF_6$ in ethylene carbonate (EC): dimethyl carbonate (DMC): ethyl methyl carbonate (EMC) (1:1:1, v:v:v)) (DodoChem, China, www.dodoChem.com), LiDFOB additive (DodoChem, China, 99.9%), Super P (Alfa Aesar), poly(vinylidene) fluoride (PVDF, Aladdin, 99.5%), N-methyl-2-pyrrolidone (NMP, Sigma-Aldrich, 99.5%, anhydrous), Al foil (Ningbo Times Aluminium Foil Technology Co., Ltd., China), carboxymethyl cellulose (CMC, Sigma-Aldrich, 99.5%), polymerized styrene butadiene rubber (SBR, Sigma-Aldrich, 99.5%), NMC811 (LG Chem), graphite (Shanghai Shanshan Tech Co., Ltd., China), lithium metal foil (Tianjin China Energy Lithium Co., Ltd., China), Cu foil (Taili Metal Foam Co., Ltd., Suzhou, China), and separator (Celgard 2400) were all used as received. More detailed information of the NMC811 and graphite materials were shown in Figure S9 and table S1.

4.2 Electrolyte preparation and cell assembly

The studied electrolyte was prepared by adding the LiDFOB additive to the baseline electrolyte with a content of 0.5, 1, 1.5, or 2 wt% in an Ar-filled glove box under stirring (H_2O and HF contents in the glove box were controlled below 20 and 50 ppm, respectively). The positive electrodes were prepared by coating a slurry of the NMC811, conductive carbon (Super P), and PVDF (9:0.5:0.5, w/w/w) in NMP on an Al foil. After drying, the electrode was cut into round disks with active material loading of ~6 mg cm^{-2} (diameter: 12 mm, the areal capacity: ~1.2 mAh cm^{-2}). The graphite negative electrode was fabricated by coating a slurry with graphite, CMC, Super-P, and SBR in a weight ratio of 93:2:1.5:2.5 on a Cu current collector (diameter: 14 mm, the areal capacity: ~1.26 mAh cm^{-2}). The N/P ratio (N: negative electrode capacity, P: positive electrode capacity) was carefully controlled at ≈1.05. The positive electrode must be prepared under dew point lower than -50 ℃. All batteries were in coin cell type and were fabricated using Celgard 2400 microporous membranes as separator and ~80 μL electrolyte inside an Ar-filled glove box. Ten cells were assembled for each electrolyte group, and five cells with consistent OCV for each electrolyte group were chosen for further electrochemical performance evaluation.

4.3 Electrochemical performance measurement

The stability of electrolytes with various compositions were obtained by means of LSV measurement on an electrochemical workstation at a scan rate of 1 mV s^{-1} (VMP300, Bio-Logic). Electrochemical impedance spectroscopy (EIS) measurements were conducted at frequencies of 5 MHz to 0.01 Hz using a voltage amplitude of 10 mV (VMP300, Bio-Logic). Long-term cycling tests were conducted on a battery analyser (Neware, Shenzhen Neware Co., Ltd.). For Li metal cells, two activation cycles were first conducted at 0.1 C before constant current-constant voltage charge/discharge cycling at C/3 and 1C between 4.4–2.8 V. For full cells, two activation cycles were first conducted at 0.1 C before constant current-constant voltage charge/discharge cycling at 1/3 C between 4.3–2.8 V.

4.4 Characterization

XRD patterns were collected on a Bruker D8 Advance (Germany) diffractometer at a scan rate of 1 ° min^{-1} using Cu Kα radiation (λ=0.154 nm). Topas 6 was used for XRD Rietveld refinement; Morphology characterizations were conducted using field-emission transmission electron microscopy (FE-TEM; Tecnai G2 F20 S-Twin) and field-emission scanning electron microscopy (FE-SEM; Japan HITACHI S4800), and all the tested samples were washed using dimethyl carbonate (DMC, battery grade, DodoChem, China) before the measurement; ICP-MS was executed using Thermo Fisher Scientific iCAP QC. XPS measurements were conducted on a PHI-5000 Versaprobe instrument employing Al Kα radiation with an analysis area of each sample is around 100 μm × 100 μm. All the tested samples were in discharged state and were vacuumed overnight at 60 °C to get rid of electrolyte residues before they were transferred to the XPS instrument in a well-sealed home-made sample holder setup. The XPS data analysis were conducted using the CasaXPS software (Version 2.3.22 PR 1.0, Casa Software Ltd., U.K.) and the binding energy of the data were calibrated using the adventitious C 1s peaks (284.8 eV) as the internal standard; TOF-SIMS measurement was conducted on a TOF. SIMS 5 spectrometer (ION-TOF GmbH). A pulsed Bi_1^+ ion beam was used for the depth profiling and a Cs^+ ion beam was used for sputtering the cycled electrodes on area of 300 μm×300 μm. The GC-MS experiments were conducted on a Agilent 5977B Q-TOF instrument with a HP-5MS UI column (30 m×0.25 mm×0.25 mm) using an Agilient GC-MS Mass

Hunter for the setup control and data analysis. The cells for GC-MS measurements were assembled using a Whatman GF/D separator.

4.5 Theoretical calculations

All calculations were performed using a Materials Studio software package. Equilibrium state structures were fully optimised using the B3LYP method and the 6–311++G (d) basis set.

ASSOCIATED CONTENT

Supporting Information. Additional SEM, TEM, XPS, and elemental content data of various electrodes (PDF) are available free of charge at XXX.

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Author Contributions

Y.S. conceived the original idea. Q.D., Y.S., and L.C. designed all the experiments. Q.D. conducted the experiments. Q.D., Y.S., and L.C. wrote the paper. All authors were involved in analysis of the experimental data.

Notes

The authors declare no competing financial interest

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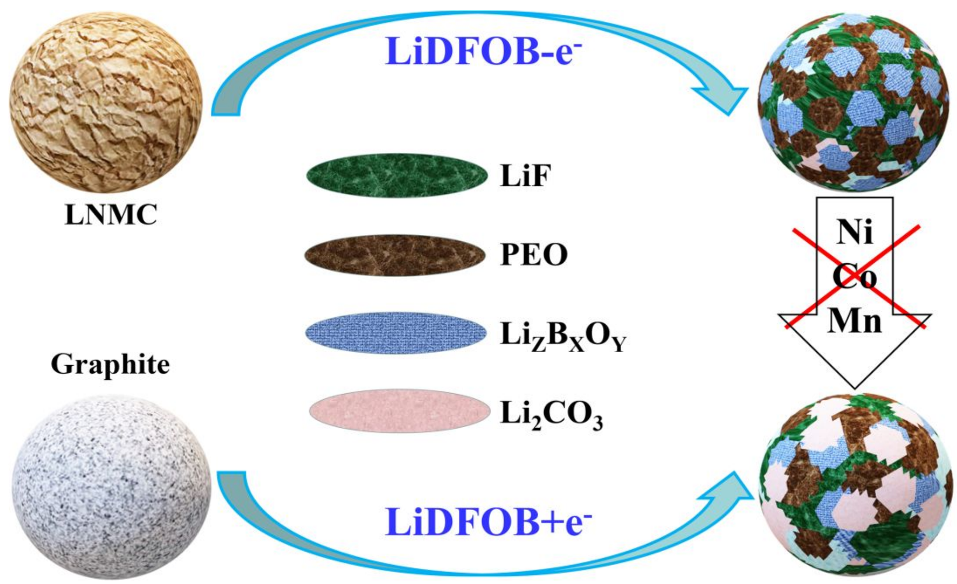

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